Figure 9.1. Testing Bell's theorem in the laboratory.

Actually performing the experiment fell to Clauser. He had been hired as a postdoc at Berkeley to do radio astronomy with Charles Townes, an astrophysicist who had won the Nobel Prize several years earlier for inventing the laser. When Clauser arrived, he told Townes about his hopes to adapt the Kocher-Commins experiment—which had been done at Berkeley—to perform a test of Bell's theorem. "I said, 'Hey look, I've got this great experiment I want to do,'" Clauser recalled, "and [Townes] goes, 'Well, why don't you give my group a seminar and tell me how it's all gonna work. And in the meantime, we'll haul up Gene Commins.'" So Clauser gave a talk explaining Bell's inequality and how the Kocher-Commins experiment could be modified to test it, hoping to get Townes interested and persuade Commins to loan him his experimental equipment. But Commins was thoroughly unimpressed by Clauser's talk. He had originally envisioned his experiment with Kocher as a mere classroom demonstration, not as a real test of EPR. That experiment had ended up being far more difficult and time-consuming than he had anticipated, and the last thing he wanted to do was to sink more time and money into a project that he saw as pointless. "Commins thought it was a total crock," Clauser said. Thankfully, Townes disagreed. "Townes thought 'Y'know, this is an interesting experiment.' And without that, I would have been dead.… At the end of [my seminar], Townes kind of puts his arm around Gene Commins and says, 'Well, what do you think of this, Gene?It looks like a very interesting experiment to me.'" Townes managed to convince the reluctant Commins to loan Clauser the equipment, to split the cost of the experiment—and to lend Clauser a graduate student from Commins's group, Stuart Freedman. Clauser and Freedman spent the next two years scrounging up the rest of the equipment they'd need for the experiment —"I've gotten pretty good at dumpster diving," Clauser boasted later— including an old telephone relay that they repurposed to control the motions of the polarizers. Once the equipment was assembled and tested, Clauser and Freedman assiduously collected two hundred hours' worth of data. Finally, in 1972, Clauser and Freedman published their results: quantum mechanics had survived. Bell's inequality was violated—and something awfully strange was going on in nature.